UTKARSH LAL

Data Scientist | previously **@ Amazon, American Express, and Accenture | MSBA @ UCLA Anderson | 3 YOE in Spark, ML, LLMs, Time Series**
utkarshlal@ucla.edu LinkedIn | Portfolio | Github **|** Google Scholar +1 (424)-467-2478

EDUCATION

UCLA ANDERSON Los Angeles, CA
MS in Business Analytics (MSBA) (Merit Scholarship Recipient) Sep 2024 - Expected Dec 2025
Machine Learning, Optimization with Python, Competitive Analytics, Prescriptive Analytics, Stats in R, Data Management.

MASSACHUSETTS INSTITUTE OF TECHNOLOGY Online
Applied Data Science Program: Recommender Engines, Deep Neural Networks, Natural Language Processing Jun 2022 – Oct 2022

MANIPAL INSTITUTE OF TECHNOLOGY KA, India
B.Tech Computer Science and Engineering (Specialization in Big Data) Jul 2017 – July 2021
Machine Learning with Big Data, Distributed Computing in Spark, Object Oriented Programming, Data Structures, Exploratory Data Analysis

TECHNICAL SKILLS AND CERTIFICATIONS

Languages: Python, PySpark, SQL, R ; **Databases** : Oracle,Cassandra, MySQL, MongoDB ; **BI Tools** : Alteryx, Excel, Tableau, QuickSight, Power BI
Frameworks: Scikit-learn, Pandas, PyTorch, Keras, NLP: (NLTK, Transformers), Matplotlib, pmdarima, Flask, Docker, Git
Cloud: AWS Sagemaker, Lambda, Databricks, GCP BigQuery, Kubernetes, Vertex AI, DataProc, Cloud Functions
Certifications: Google Cloud Certified: Professional Data Engineer, Databricks Developer Foundations, Accenture Data Engineering

PROFESSIONAL EXPERIENCE

AMAZON Noida, India
Data Engineer Mar 2024 – Sep 2024

- Developed and monitored data pipelines using Spark SQL and PySpark for Amazon Global Store, managing export-import data arcs from North America and Europe, handling petabytes of data, and **reducing cloud-compute expenditure by 50%**.
- Senior engineer in ‘Project Mariner’, a complex migration initiative for transforming 200+ ETL pipelines using **Apache Spark**. Employed extensive spark optimization, reducing average execution cost to 0.35$ per day for each ETL job (**70% reduction**).
- Automated the backfilling of 100+ TB of data for newly constructed architecture using Spark, **reducing development time by 50+ hours**.
- Built a POC **time series forecasting** model for predicting sales revenue for Amazon Global Store’s new catalog leveraging **RNNs**.
- Built **Spark Streaming** pipelines for clickstream on Databricks, processing over 20,000 records per minute for an external client.
- Served **100 users/minute** by implementing a QuickSight Dashboard for real-time monitoring of ETL job execution metrics and disk utilization across 5 Redshift clusters and 4 Amazon proprietary DW platforms, preventing weeks of access approval delays.

STATUSNEO (Client: Reliance Jio AI Center of Excellence) Gurugram, India
Data Science Consultant Aug 2022 – Dec 2023

- Product Popularity Forecaster**:** Trained **SARIMAX** models to forecast sales revenue and quantity of an e-retail platform called JioMart, uncovering city-wise sales trends and seasonality. Created an exogenous feature, enhancing model accuracy, and **improving forecasts by 10%**.
- Product Popularity Calculator: Leveraged **Huber** and XGB Regressors coupled with custom exponential smoothing to quantify product popularity for the Tira catalog, based on user behaviour and the feature importance of click events, improving search recommendations.
- Fine-tuned **DistilBERT** (**QLoRA**) to create sentiment inferencing APIs for Tira reviews/ratings, **reducing latency by 25%.**
- Constructed an **Autosuggest REST API** for JioTV to recommend content based on trending searches and clicks, in under 3 seconds. Developed a high-throughput Pyspark pipeline handling TBs of data across **Google Cloud Storage**, **Cassandra**, and a Redis cache layer.

AMERICAN EXPRESS Gurugram, India
Data Science Analyst – Product Development Mar 2022 – Jun 2022

- Worked in the Credit and Fraud Risk Capabilities team. Engineered PySpark pipelines for Expected Credit Loss calculations. Enhanced Fraud Detection Models using Random Forest and XGBoost by feature engineering, and **Principal Component Analysis**, decreasing **run time by 50%**.

ACCENTURE Gurugram, India
Data Engineering Analyst Aug 2021 – Mar 2022

- Built a **churn prediction** model using Call Detail Record data of a telecom organization, achieving **90% average accuracy**. Constructed batch processing data pipelines in Google Cloud Functions and **DataProc**. Orchestrated pipelines using Apache **Airflow** and Cloud Composer.

KIMBERLY-CLARK Bangalore, India
Agile Analytics Engineer Intern Jan 2021 – Jul 2021

- Scraped product reviews from Amazon and performed **Topic Modelling**. Utilized TF-IDF and **Latent Dirichlet Allocation**, identifying the common features in negative reviews. Visualized Topics using the PyLDAvis and evaluated the model using Coherence and Perplexity metrics.
- Built and orchestrated ETL pipelines in Alteryx and KNIME, integrating data from various data sources and created a dashboard in Quicksight.

SELECTED RESEARCH PUBLICATIONS AND PROJECTS

1. Fractal dimensions and machine learning for detection of Parkinson’s disease in resting-state electroencephalography**:** Time Series Decomposition combined with ML yielded accuracies >95%. Published in ***Neural Computing and Applications journal*** **(Q1; IF: 6.0)**.
2. Temporal Feature Extraction and Machine Learning for Classification of Sleep Stages Using Telemetry Polysomnography**:** ML model achieved unprecedented accuracies of up to 97%, surpassing all DNNs published previously. Published in the ***Brain Sciences*** **journal (Q2; IF: 3.0)**.
3. https://doi.org/10.3390/brainsci14040335: Leveraged Singular Value Decomposition Entropy and Explainable AI to differentiate between Alzheimer’s Disease and Frontotemporal Dementia with over 93% average accuracy**.** ***Brain Sciences*** **(Q2; IF: 3.0)**.
4. HealthCharge Classifier @ **Accenture**: Employed **LightGBM, TF-IDF**, and **cosine similarity** for Automated Expense Categorization for Healthcare Billing by constructing a corpus of 100K keywords and implementing a quick inferencing module. Containerized using **Docker** and **AWS EKS**.
5. **Facial Emotion Recognition using CNN @ MIT**: Crafted a **Tensorflow**-based CNN to classify 4 emotion states (happy, sad, neutral, surprised) by fine-tuning and comparing VGG16, ResNet, and EfficientNet models.